and tested?" A Washington, DC, native who graduated from the US Naval Academy, Haney is a quiet, lanky submariner who commanded US Pacific Fleet before assuming control of America's nuclear arsenal. "We have a culture of asking, 'What aren't we doing right?' to try and avoid mistakes," Haney said. He then paused and asked, "Will North Korea make shortcuts in a very expensive enterprise?"[2]

Nuclear weapons command and safety is a far less sexy topic than nuclear warfighting, though senior US officials such as former secretary of state George Shultz and secretaries of defense William J. Perry and Ash Carter have spent decades exploring nuclear stability and warning about the dangers of complacency.[3] It was the specter of inevitable nuclear mistakes and accidents that spawned the greatest nightmares of the Cold War, dystopic visions in books and movies such as *Fail Safe* and *Dr. Strangelove* of a world incinerated by an atomic fireball (or nearly so) due to a madman, a blown fuse, a garbled message, or a simple computer game. And the public had good reason to worry, which is why the specter of a North Korean nuclear arsenal should terrify even those who firmly believe Kim is not so suicidal as to launch an unprovoked nuclear attack. In a North Korean nuclear world, we have to be sure not only of Kim's rationality but also that the command and control system he puts into place will not break down due to mechanical or human error, that his weapons won't explode due to a glitch or poor maintenance and won't be launched accidentally. Yet when dealing with nuclear surety in North Korea, we enter a world composed almost entirely of "known unknowns."

III.

The risks of a nuclear program start with the very act of building a bomb. At the dawn of the atomic age that process was often shockingly lax compared to what it became a few decades later. Though no one knows for sure, it is unlikely that North Korean scientists work in the primitive conditions in which two American scientists died from